

# OC Core Reviewer Attack and Response Map v1.3.3

Ontology of Continua Core release review artifact

Alexander Yashin

Version 1.3.3

## Contents

|          |                                                            |           |
|----------|------------------------------------------------------------|-----------|
| <b>1</b> | <b>OC Core Reviewer Attack and Response Map v1.3.3</b>     | <b>10</b> |
| 1.1      | Publication Identity . . . . .                             | 10        |
| 1.2      | Dedication . . . . .                                       | 11        |
| 1.3      | Acknowledgements . . . . .                                 | 11        |
| 1.4      | Abstract . . . . .                                         | 11        |
| 1.5      | Reader Contract . . . . .                                  | 11        |
| 1.6      | Table of Contents . . . . .                                | 11        |
| <b>2</b> | <b>Body</b>                                                | <b>12</b> |
| 2.1      | Block 2 . . . . .                                          | 12        |
| 2.1.1    | Define Reader Contract . . . . .                           | 12        |
| 2.1.2    | Define Intended Audiences . . . . .                        | 13        |
| 2.1.3    | Define What OC Core Claims . . . . .                       | 14        |
| 2.1.4    | Define What OC Core Does Not Claim . . . . .               | 14        |
| 2.1.5    | Define Release Scope vs Full Science Program . . . . .     | 15        |
| 2.1.6    | Define Claim Promotion, Demotion, . . . . .                | 16        |
| 2.2      | Block 13 . . . . .                                         | 16        |
| 2.2.1    | Define Falsifier Registry . . . . .                        | 16        |
| 2.2.2    | Define Counterexample Search . . . . .                     | 17        |
| 2.2.3    | Define Known Failure Modes . . . . .                       | 18        |
| 2.2.4    | Define Unsupported Strong Claims . . . . .                 | 18        |
| 2.2.5    | Define Demotion Rules . . . . .                            | 19        |
| 2.2.6    | Define Residual Scientific Risks . . . . .                 | 20        |
| 2.2.7    | Define Full-Science Background Obligations . . . . .       | 20        |
| 2.3      | Block 14 . . . . .                                         | 21        |
| 2.3.1    | Define Novelty Criteria . . . . .                          | 21        |
| 2.3.2    | Define Non-Equivalence Criteria . . . . .                  | 22        |
| 2.3.3    | Define Overlap With Prior Art . . . . .                    | 22        |
| 2.3.4    | Define Residual-Delta Analysis . . . . .                   | 23        |
| 2.3.5    | Define Competitor-by-Competitor Comparison . . . . .       | 24        |
| 2.3.6    | Define Response to “This Is Just Another Theory” . . . . . | 24        |
| 2.3.7    | Define Positioning of OC Core . . . . .                    | 25        |
| 2.4      | Block 16 . . . . .                                         | 26        |

|        |                                                            |    |
|--------|------------------------------------------------------------|----|
| 2.4.1  | Define Review Protocol . . . . .                           | 26 |
| 2.4.2  | Define Cerberus Role Map . . . . .                         | 27 |
| 2.4.3  | Define Critical . . . . .                                  | 27 |
| 2.4.4  | Define Closure Evidence . . . . .                          | 28 |
| 2.4.5  | Define Claim Corrections . . . . .                         | 29 |
| 2.4.6  | Define Hostile Reviewer Objections . . . . .               | 29 |
| 2.4.7  | Define Response Matrix . . . . .                           | 30 |
| 2.4.8  | Define Remaining Non-Blocking Caveats . . . . .            | 31 |
| 2.5    | Block 18 . . . . .                                         | 31 |
| 2.5.1  | Define Versioning . . . . .                                | 31 |
| 2.5.2  | Define Public Release Asset Set . . . . .                  | 32 |
| 2.5.3  | Define Journal Package Map . . . . .                       | 33 |
| 2.5.4  | Define Citation . . . . .                                  | 33 |
| 2.5.5  | Define Publication Boundary . . . . .                      | 34 |
| 2.5.6  | Define Owner Approval Boundary . . . . .                   | 35 |
| 2.5.7  | Define Incident . . . . .                                  | 35 |
| 2.5.8  | Define External Use Guidance . . . . .                     | 36 |
| 2.6    | Block 999 . . . . .                                        | 37 |
| 2.6.1  | Adversarial Review Structure . . . . .                     | 37 |
| 2.6.2  | Limits of recovery . . . . .                               | 37 |
| 2.6.3  | K-level criteria . . . . .                                 | 37 |
| 2.6.4  | Falsifiability of K_0K_0 . . . . .                         | 37 |
| 2.6.5  | Fundamental Falsifiability Principles for K_0K_0 . . . . . | 37 |
| 2.6.6  | Structural Falsifiability Conditions . . . . .             | 38 |
| 2.6.7  | Boundary . . . . .                                         | 38 |
| 2.6.8  | Energy Functional . . . . .                                | 38 |
| 2.6.9  | Cycle . . . . .                                            | 38 |
| 2.6.10 | Falsifiability via Modal Spaces . . . . .                  | 38 |
| 2.6.11 | Falsifiability via Meta-Cycles C_10 . . . . .              | 39 |
| 2.6.12 | Falsifiability via Cycles C_11 . . . . .                   | 39 |
| 2.6.13 | Falsifiability via Universal Cycles C_12 . . . . .         | 39 |
| 2.6.14 | Percolation . . . . .                                      | 39 |
| 2.6.15 | Boundary . . . . .                                         | 39 |
| 2.6.16 | Falsifiability via Chemical Potentials . . . . .           | 39 |
| 2.6.17 | Falsifiability via Cycles C_4C_4 . . . . .                 | 40 |
| 2.6.18 | Falsifiability via Membrane . . . . .                      | 40 |
| 2.6.19 | Falsifiability via Neural Cycles C_5C_5 . . . . .          | 40 |
| 2.6.20 | Falsifiability via Cognitive Cycles C_6C_6 . . . . .       | 40 |
| 2.6.21 | Falsifiability via Social Cycles C_7C_7 . . . . .          | 40 |
| 2.6.22 | Falsifiability via Civilisational Cycles C_8C_8 . . . . .  | 41 |
| 2.6.23 | Falsifiability via Scientific Cycles C_9C_9 . . . . .      | 41 |
| 2.6.24 | Falsifiability via Dynamics . . . . .                      | 41 |
| 2.6.25 | Falsifier . . . . .                                        | 41 |
| 2.6.26 | Theorem fate correction after external review . . . . .    | 41 |
| 2.6.27 | Worked Examples, Counterexamples, . . . . .                | 41 |
| 2.6.28 | What would count as a real counterexample . . . . .        | 42 |
| 2.6.29 | Mathematics: Strict closure . . . . .                      | 42 |
| 2.6.30 | Simulation . . . . .                                       | 42 |

|        |                                                                         |    |
|--------|-------------------------------------------------------------------------|----|
| 2.6.31 | Reviewer challenge 3: Minimality Component Witness . . . . .            | 42 |
| 2.6.32 | Reviewer challenge 4: Kleavel Transition Witness . . . . .              | 43 |
| 2.6.33 | Falsifiability via Symbolic Systems . . . . .                           | 43 |
| 2.6.34 | Falsifiability via Knowledge Systems . . . . .                          | 43 |
| 2.6.35 | (P10.11) Completeness Is Structurally Limited . . . . .                 | 43 |
| 2.6.36 | (i) Structural falsifiability . . . . .                                 | 43 |
| 2.6.37 | Replay Matrix for Independent Review . . . . .                          | 44 |
| 2.6.38 | Reviewer challenge 5: Numeric Replay Quarantine . . . . .               | 44 |
| 2.6.39 | Attack Family 3: The Empirical Rows Are Too Weak . . . . .              | 44 |
| 2.6.40 | Formal-Mathematics Reviewer . . . . .                                   | 44 |
| 2.6.41 | Empirical-Statistics Reviewer . . . . .                                 | 44 |
| 2.6.42 | Publication-Metadata Reviewer . . . . .                                 | 45 |
| 2.6.43 | Worked Attack Transcript B: Replay Surface . . . . .                    | 45 |
| 2.6.44 | Reviewer challenge 7: Prior Art Positioning . . . . .                   | 45 |
| 2.6.45 | Worked Attack Transcript C: Prior-Art Surface . . . . .                 | 45 |
| 2.6.46 | Phenomenon Coverage . . . . .                                           | 45 |
| 2.6.47 | Reviewer challenge 6: Phenomenon Coverage Model Card . . . . .          | 46 |
| 2.6.48 | Reviewer challenge 8: Attack Matrix Distinct Surface Coverage . . . . . | 46 |
| 2.6.49 | Attack Family 4: Phenomenon Coverage Is Inflated . . . . .              | 46 |
| 2.6.50 | Visual Route - Reviewer Route Figures . . . . .                         | 46 |
| 2.6.51 | Reviewer Navigation Matrix . . . . .                                    | 46 |
| 2.6.52 | Review questions, compact route, . . . . .                              | 47 |
| 2.6.53 | Audit, provenance, . . . . .                                            | 47 |
| 2.6.54 | Compact reading order by reviewer role . . . . .                        | 47 |
| 2.6.55 | Rapid critical reviewer . . . . .                                       | 47 |
| 2.6.56 | Formal mathematical reviewer . . . . .                                  | 47 |
| 2.6.57 | Generalist scientific reviewer . . . . .                                | 47 |
| 2.6.58 | Reviewer note . . . . .                                                 | 48 |
| 2.6.59 | Tuple . . . . .                                                         | 48 |
| 2.6.60 | Reviewer acknowledgement . . . . .                                      | 48 |
| 2.6.61 | Hostile-review backlog . . . . .                                        | 48 |
| 2.6.62 | Shortest hostile-review checklist . . . . .                             | 48 |
| 2.6.63 | Reviewer Entry Points . . . . .                                         | 49 |
| 2.6.64 | Does OC actually explain minimality versus relabeling attack? . . . . . | 49 |
| 2.6.65 | Reviewer Checklist . . . . .                                            | 49 |
| 2.6.66 | Reviewer Burden . . . . .                                               | 49 |
| 2.6.67 | Reviewer Response Method . . . . .                                      | 49 |
| 2.6.68 | Claim Boundary . . . . .                                                | 49 |
| 2.6.69 | Reviewer challenge 1: Theorem Proof Binding . . . . .                   | 50 |
| 2.6.70 | Reviewer challenge 2: Theorem Public Surface Binding . . . . .          | 50 |
| 2.6.71 | Adversarial Review Method . . . . .                                     | 50 |
| 2.6.72 | Attack Family 1: The Theory Is Only a Reframing . . . . .               | 50 |
| 2.6.73 | Attack Family 2: The Formal Claims Are Theatre . . . . .                | 50 |
| 2.6.74 | Attack Family 5: Public Wording Outruns Evidence . . . . .              | 51 |
| 2.6.75 | Attack Family 6: The Package Is Not a Scientific Text . . . . .         | 51 |
| 2.6.76 | Reviewer Role Playbook . . . . .                                        | 51 |
| 2.6.77 | Computational-Semantics Reviewer . . . . .                              | 51 |
| 2.6.78 | Release-Engineering Reviewer . . . . .                                  | 51 |

|         |                                                                  |    |
|---------|------------------------------------------------------------------|----|
| 2.6.79  | Synthesis Reviewer . . . . .                                     | 52 |
| 2.6.80  | Minimum Hostile Review Walkthrough . . . . .                     | 52 |
| 2.6.81  | End-to-End Attack Drill . . . . .                                | 52 |
| 2.6.82  | Cheap Review Cascade . . . . .                                   | 52 |
| 2.6.83  | Worked Attack Transcript A: Theorem Surface . . . . .            | 52 |
| 2.6.84  | Worked Attack Transcript D: Public Surface . . . . .             | 52 |
| 2.6.85  | Worked Attack Transcript E: Synthesis Surface . . . . .          | 53 |
| 2.6.86  | Falsifiability via Institutions . . . . .                        | 53 |
| 2.6.87  | Limitations of the current formalism . . . . .                   | 53 |
| 2.6.88  | Limits of recovery . . . . .                                     | 53 |
| 2.6.89  | Extended Falsifiability . . . . .                                | 53 |
| 2.6.90  | Meaning of Falsifiability for Structural Theories . . . . .      | 54 |
| 2.6.91  | Structural vs phenomenological falsifiability . . . . .          | 54 |
| 2.6.92  | Structural falsification channels . . . . .                      | 54 |
| 2.6.93  | Cross-domain falsification logic . . . . .                       | 54 |
| 2.6.94  | Falsification criteria . . . . .                                 | 54 |
| 2.6.95  | Source-tradition remediation after external review . . . . .     | 54 |
| 2.6.96  | K-level criteria . . . . .                                       | 55 |
| 2.6.97  | Theorem fate correction after external review . . . . .          | 55 |
| 2.6.98  | Meta-limit extension . . . . .                                   | 55 |
| 2.6.99  | Type~VII Experiments: Detectability . . . . .                    | 55 |
| 2.6.100 | Relation to Falsifiability . . . . .                             | 55 |
| 2.6.101 | Falsifiability of K_0K_0 . . . . .                               | 56 |
| 2.6.102 | Fundamental Falsifiability Principles for K_0K_0 . . . . .       | 56 |
| 2.6.103 | Structural Falsifiability Conditions . . . . .                   | 56 |
| 2.6.104 | Boundary . . . . .                                               | 56 |
| 2.6.105 | Falsifiability of K_1K_1 . . . . .                               | 56 |
| 2.6.106 | Structural . . . . .                                             | 56 |
| 2.6.107 | Energy Functional . . . . .                                      | 57 |
| 2.6.108 | Cycle . . . . .                                                  | 57 |
| 2.6.109 | Embedding-Space . . . . .                                        | 57 |
| 2.6.110 | Falsifiability of K_10 . . . . .                                 | 57 |
| 2.6.111 | Falsifiability via Reflexive . . . . .                           | 57 |
| 2.6.112 | Falsifiability via Category-Theoretical . . . . .                | 58 |
| 2.6.113 | Falsifiability via Modal Spaces . . . . .                        | 58 |
| 2.6.114 | Falsifiability via Meta-Level Potentials P_10 . . . . .          | 58 |
| 2.6.115 | Falsifiability via Meta-Flows $\hat{J}(10)$ . . . . .            | 58 |
| 2.6.116 | Falsifiability via Meta-Cycles C_10 . . . . .                    | 58 |
| 2.6.117 | Falsifiability via Structural Tension T_10 . . . . .             | 59 |
| 2.6.118 | Falsifiability of the Transition K_9 K_10 . . . . .              | 59 |
| 2.6.119 | Falsifiability of K_11 . . . . .                                 | 59 |
| 2.6.120 | Falsifiability via Meta-Meta Coherence . . . . .                 | 59 |
| 2.6.121 | Falsifiability via Higher-Order Categorical Structures . . . . . | 59 |
| 2.6.122 | Falsifiability via Higher-Order Modal Spaces . . . . .           | 59 |
| 2.6.123 | Falsifiability via Potentials P_11 . . . . .                     | 60 |
| 2.6.124 | Falsifiability via Flows $\hat{J}(11)$ . . . . .                 | 60 |
| 2.6.125 | Falsifiability via Cycles C_11 . . . . .                         | 60 |
| 2.6.126 | Falsifiability via Structural Tension T_11 . . . . .             | 60 |

|                                                              |    |
|--------------------------------------------------------------|----|
| 2.6.127 Falsifiability of the Transition $K_{10} K_{11}$     | 60 |
| 2.6.128 Falsifiability of $K_{12}$                           | 61 |
| 2.6.129 Falsifiability via Universal Structural Consistency  | 61 |
| 2.6.130 Falsifiability via Universal Modal                   | 61 |
| 2.6.131 Falsifiability via Universal Potentials $P_{12}$     | 61 |
| 2.6.132 Falsifiability via Universal Flows $J^{\wedge}(12)$  | 61 |
| 2.6.133 Falsifiability via Universal Cycles $C_{12}$         | 61 |
| 2.6.134 Falsifiability via Universal Boundaries ( $K_{12}$ ) | 62 |
| 2.6.135 Falsifiability of $K_{11} K_{12}$ Transition         | 62 |
| 2.6.136 Falsifiability of $K_2 K_2$                          | 62 |
| 2.6.137 Topological                                          | 62 |
| 2.6.138 Percolation                                          | 62 |
| 2.6.139 Boundary                                             | 63 |
| 2.6.140 Potentials, Flows                                    | 63 |
| 2.6.141 Dynamic                                              | 63 |
| 2.6.142 Transition                                           | 63 |
| 2.6.143 Falsifiability of $K_3 K_3$                          | 63 |
| 2.6.144 Falsifiability via Chemical Configuration            | 63 |
| 2.6.145 Falsifiability via Reaction Networks                 | 64 |
| 2.6.146 Falsifiability via RAF Networks                      | 64 |
| 2.6.147 Falsifiability via Chemical Potentials               | 64 |
| 2.6.148 Falsifiability via Boundaries                        | 64 |
| 2.6.149 Falsifiability via Dynamics                          | 64 |
| 2.6.150 Falsifiability via Transition $K_2 K_3 K_2 K_3$      | 65 |
| 2.6.151 Falsifiability of $K_4 K_4$                          | 65 |
| 2.6.152 Falsifiability via Compartment Formation             | 65 |
| 2.6.153 Falsifiability via Biological Potentials $P_4 P_4$   | 65 |
| 2.6.154 Falsifiability via Flows $J_4 J_4$                   | 65 |
| 2.6.155 Falsifiability via Cycles $C_4 C_4$                  | 65 |
| 2.6.156 Falsifiability via Membrane-Bound Information        | 66 |
| 2.6.157 Falsifiability via Structural Tension $T_4 T_4$      | 66 |
| 2.6.158 Falsifiability via Transition $K_3 K_4 K_3 K_4$      | 66 |
| 2.6.159 Falsifiability of $K_5 K_5$                          | 66 |
| 2.6.160 Falsifiability via Membrane                          | 66 |
| 2.6.161 Falsifiability via Electrical Potentials $P_5 P_5$   | 67 |
| 2.6.162 Falsifiability via Ion Flows $J_5 J_5$               | 67 |
| 2.6.163 Falsifiability via Neural Cycles $C_5 C_5$           | 67 |
| 2.6.164 Falsifiability via Stochastic Logic                  | 67 |
| 2.6.165 Falsifiability via Structural Tension $T_5 T_5$      | 67 |
| 2.6.166 Falsifiability of Transition $K_4 K_5 K_4 K_5$       | 67 |
| 2.6.167 Falsifiability of $K_6 K_6$                          | 68 |
| 2.6.168 Falsifiability via Representational Structure        | 68 |
| 2.6.169 Falsifiability via Binding Dynamics                  | 68 |
| 2.6.170 Falsifiability via Predictive Processing             | 68 |
| 2.6.171 Falsifiability via Comparison                        | 68 |
| 2.6.172 Falsifiability via Memory                            | 69 |
| 2.6.173 Falsifiability via Cognitive Flows $J_6 J_6$         | 69 |
| 2.6.174 Falsifiability via Cognitive Cycles $C_6 C_6$        | 69 |

|                                                                                        |    |
|----------------------------------------------------------------------------------------|----|
| 2.6.175 Falsifiability via Structural Tension T_6T_6 . . . . .                         | 69 |
| 2.6.176 Falsifiability of Transition K_5 K_6K_5 K_6 . . . . .                          | 69 |
| 2.6.177 Falsifiability of K_7K_7 . . . . .                                             | 69 |
| 2.6.178 Falsifiability via Communication Structure . . . . .                           | 70 |
| 2.6.179 Falsifiability via Trust . . . . .                                             | 70 |
| 2.6.180 Falsifiability via Roles . . . . .                                             | 70 |
| 2.6.181 Falsifiability via Coordination Dynamics . . . . .                             | 70 |
| 2.6.182 Falsifiability via Resource Flows J_7J_7 . . . . .                             | 70 |
| 2.6.183 Falsifiability via Social Cycles C_7C_7 . . . . .                              | 71 |
| 2.6.184 Falsifiability via Structural Tension T_7T_7 . . . . .                         | 71 |
| 2.6.185 Falsifiability of Transition K_6 K_7K_6 K_7 . . . . .                          | 71 |
| 2.6.186 Falsifiability of K_8K_8 . . . . .                                             | 71 |
| 2.6.187 Falsifiability via Technology . . . . .                                        | 71 |
| 2.6.188 Falsifiability via Cultural Dynamics . . . . .                                 | 71 |
| 2.6.189 Falsifiability via Civilisational Flows J_8J_8 . . . . .                       | 72 |
| 2.6.190 Falsifiability via Civilisational Cycles C_8C_8 . . . . .                      | 72 |
| 2.6.191 Falsifiability via Structural Tension T_8T_8 . . . . .                         | 72 |
| 2.6.192 Falsifiability of Transition K_7 K_8K_7 K_8 . . . . .                          | 72 |
| 2.6.193 Falsifiability of K_9K_9 . . . . .                                             | 72 |
| 2.6.194 Falsifiability via Coherence . . . . .                                         | 73 |
| 2.6.195 Falsifiability via Paradigm Dynamics . . . . .                                 | 73 |
| 2.6.196 Falsifiability via Epistemic Potentials . . . . .                              | 73 |
| 2.6.197 Falsifiability via Methodological Structure . . . . .                          | 73 |
| 2.6.198 Falsifiability via Scientific Flows J_9J_9 . . . . .                           | 73 |
| 2.6.199 Falsifiability via Scientific Cycles C_9C_9 . . . . .                          | 73 |
| 2.6.200 Falsifiability via Structural Tension T_9T_9 . . . . .                         | 74 |
| 2.6.201 Falsifiability of the Transition K_8 K_9K_8 K_9 . . . . .                      | 74 |
| 2.6.202 Falsifiability . . . . .                                                       | 74 |
| 2.6.203 Structural Meaning of Falsifiability . . . . .                                 | 74 |
| 2.6.204 Universal Falsifiability Criteria . . . . .                                    | 74 |
| 2.6.205 Cross-Level Falsifiability Structure . . . . .                                 | 75 |
| 2.6.206 Falsifiability via Dynamics . . . . .                                          | 75 |
| 2.6.207 Falsifiability via Embedding Spaces . . . . .                                  | 75 |
| 2.6.208 Falsifiability in Practice: Multi-Domain Implementation . . . . .              | 75 |
| 2.6.209 Falsifier . . . . .                                                            | 75 |
| 2.6.210 Claim Boundary . . . . .                                                       | 75 |
| 2.6.211 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-FALSE-<br>CLAIM-FALSE . . . . . | 76 |
| 2.6.212 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-FALSE-<br>CLAIM-TRUE . . . . .  | 76 |
| 2.6.213 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-TRUE-<br>CLAIM-FALSE . . . . .  | 76 |
| 2.6.214 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-TRUE-<br>CLAIM-TRUE . . . . .   | 76 |
| 2.6.215 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-FALSE-<br>CLAIM-FALSE . . . . .  | 77 |
| 2.6.216 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-FALSE-<br>CLAIM-TRUE . . . . .   | 77 |

|                                                                                   |    |
|-----------------------------------------------------------------------------------|----|
| 2.6.217 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-FALSE . . . . .  | 77 |
| 2.6.218 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-TRUE . . . . .   | 77 |
| 2.6.219 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-FALSE . . . . . | 77 |
| 2.6.220 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-TRUE . . . . .  | 78 |
| 2.6.221 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-FALSE . . . . .  | 78 |
| 2.6.222 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-TRUE . . . . .   | 78 |
| 2.6.223 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-FALSE . . . . .  | 78 |
| 2.6.224 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-TRUE . . . . .   | 79 |
| 2.6.225 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-FALSE . . . . .   | 79 |
| 2.6.226 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-FALSE . . . . . | 79 |
| 2.6.227 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-TRUE . . . . .  | 79 |
| 2.6.228 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-FALSE . . . . .  | 79 |
| 2.6.229 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-TRUE . . . . .   | 80 |
| 2.6.230 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-FALSE . . . . .  | 80 |
| 2.6.231 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-TRUE . . . . .   | 80 |
| 2.6.232 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-FALSE . . . . .   | 80 |
| 2.6.233 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-TRUE . . . . .    | 81 |
| 2.6.234 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-FALSE . . . . .  | 81 |
| 2.6.235 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-TRUE . . . . .   | 81 |
| 2.6.236 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-FALSE . . . . .   | 81 |
| 2.6.237 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-TRUE . . . . .    | 81 |
| 2.6.238 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-FALSE . . . . .   | 82 |
| 2.6.239 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-TRUE . . . . .    | 82 |
| 2.6.240 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-FALSE . . . . .    | 82 |

|                                                                                   |    |
|-----------------------------------------------------------------------------------|----|
| 2.6.241 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-FALSE . . . . . | 82 |
| 2.6.242 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-TRUE . . . . .  | 83 |
| 2.6.243 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-FALSE . . . . .  | 83 |
| 2.6.244 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-TRUE . . . . .   | 83 |
| 2.6.245 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-FALSE . . . . .  | 83 |
| 2.6.246 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-TRUE . . . . .   | 83 |
| 2.6.247 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-FALSE . . . . .   | 84 |
| 2.6.248 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-TRUE . . . . .    | 84 |
| 2.6.249 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-FALSE . . . . .  | 84 |
| 2.6.250 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-TRUE . . . . .   | 84 |
| 2.6.251 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-FALSE . . . . .   | 85 |
| 2.6.252 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-TRUE . . . . .    | 85 |
| 2.6.253 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-FALSE . . . . .   | 85 |
| 2.6.254 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-TRUE . . . . .    | 85 |
| 2.6.255 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-FALSE . . . . .    | 85 |
| 2.6.256 PHYSICAL_REVIEW_RESEARCH . . . . .                                        | 86 |
| 2.6.257 Minimum Independent Scientific Review Packet . . . . .                    | 86 |
| 2.6.258 Limitations . . . . .                                                     | 86 |
| 2.6.259 Journal Owner-Review Packages . . . . .                                   | 86 |
| 2.6.260 Limitations of the current formalism . . . . .                            | 86 |
| 2.6.261 Extended Falsifiability . . . . .                                         | 87 |
| 2.6.262 Meaning of Falsifiability for Structural Theories . . . . .               | 87 |
| 2.6.263 Structural vs phenomenological falsifiability . . . . .                   | 87 |
| 2.6.264 Structural falsification channels . . . . .                               | 87 |
| 2.6.265 Cross-domain falsification logic . . . . .                                | 87 |
| 2.6.266 Falsification criteria . . . . .                                          | 87 |
| 2.6.267 Source-tradition remediation after external review . . . . .              | 88 |
| 2.6.268 Meta-limit extension . . . . .                                            | 88 |
| 2.6.269 Type~VII Experiments: Detectability . . . . .                             | 88 |
| 2.6.270 Relation to Falsifiability . . . . .                                      | 88 |
| 2.6.271 Falsifiability of K_1K_1 . . . . .                                        | 88 |
| 2.6.272 Structural . . . . .                                                      | 89 |
| 2.6.273 Embedding-Space . . . . .                                                 | 89 |



|                                                                |    |
|----------------------------------------------------------------|----|
| 2.6.274 Falsifiability of $K_{10}$                             | 89 |
| 2.6.275 Falsifiability via Reflexive                           | 89 |
| 2.6.276 Falsifiability via Category-Theoretical                | 89 |
| 2.6.277 Falsifiability via Meta-Level Potentials $P_{10}$      | 90 |
| 2.6.278 Falsifiability via Meta-Flows $J^{\wedge}(10)$         | 90 |
| 2.6.279 Falsifiability via Structural Tension $T_{10}$         | 90 |
| 2.6.280 Falsifiability of the Transition $K_9 K_{10}$          | 90 |
| 2.6.281 Falsifiability of $K_{11}$                             | 90 |
| 2.6.282 Falsifiability via Meta-Meta Coherence                 | 90 |
| 2.6.283 Falsifiability via Higher-Order Categorical Structures | 91 |
| 2.6.284 Falsifiability via Higher-Order Modal Spaces           | 91 |
| 2.6.285 Falsifiability via Potentials $P_{11}$                 | 91 |
| 2.6.286 Falsifiability via Flows $J^{\wedge}(11)$              | 91 |
| 2.6.287 Falsifiability via Structural Tension $T_{11}$         | 91 |
| 2.6.288 Falsifiability of the Transition $K_{10} K_{11}$       | 92 |
| 2.6.289 Falsifiability of $K_{12}$                             | 92 |
| 2.6.290 Falsifiability via Universal Structural Consistency    | 92 |
| 2.6.291 Falsifiability via Universal Modal                     | 92 |
| 2.6.292 Falsifiability via Universal Potentials $P_{12}$       | 92 |
| 2.6.293 Falsifiability via Universal Flows $J^{\wedge}(12)$    | 92 |
| 2.6.294 Falsifiability via Universal Boundaries ( $K_{12}$ )   | 93 |
| 2.6.295 Falsifiability of $K_{11} K_{12}$ Transition           | 93 |
| 2.6.296 Falsifiability of $K_2 K_2$                            | 93 |
| 2.6.297 Topological                                            | 93 |
| 2.6.298 Potentials, Flows                                      | 93 |
| 2.6.299 Dynamic                                                | 94 |
| 2.6.300 Transition                                             | 94 |
| 2.6.301 Falsifiability of $K_3 K_3$                            | 94 |
| 2.6.302 Falsifiability via Chemical Configuration              | 94 |
| 2.6.303 Falsifiability via Reaction Networks                   | 94 |
| 2.6.304 Falsifiability via RAF Networks                        | 94 |
| 2.6.305 Falsifiability via Boundaries                          | 95 |
| 2.6.306 Falsifiability via Dynamics                            | 95 |
| 2.6.307 Falsifiability via Transition $K_2 K_3 K_2 K_3$        | 95 |
| 2.6.308 Falsifiability of $K_4 K_4$                            | 95 |
| 2.6.309 Falsifiability via Compartment Formation               | 95 |
| 2.6.310 Falsifiability via Biological Potentials $P_4 P_4$     | 96 |
| 2.6.311 Falsifiability via Flows $J_4 J_4$                     | 96 |
| 2.6.312 Falsifiability via Membrane-Bound Information          | 96 |
| 2.6.313 Falsifiability via Structural Tension $T_4 T_4$        | 96 |
| 2.6.314 Falsifiability via Transition $K_3 K_4 K_3 K_4$        | 96 |
| 2.6.315 Falsifiability of $K_5 K_5$                            | 96 |
| 2.6.316 Falsifiability via Electrical Potentials $P_5 P_5$     | 97 |
| 2.6.317 Falsifiability via Ion Flows $J_5 J_5$                 | 97 |
| 2.6.318 Falsifiability via Stochastic Logic                    | 97 |
| 2.6.319 Falsifiability via Structural Tension $T_5 T_5$        | 97 |
| 2.6.320 Falsifiability of Transition $K_4 K_5 K_4 K_5$         | 97 |
| 2.6.321 Falsifiability of $K_6 K_6$                            | 98 |

|                                                                           |     |
|---------------------------------------------------------------------------|-----|
| 2.6.322 Falsifiability via Representational Structure . . . . .           | 98  |
| 2.6.323 Falsifiability via Binding Dynamics . . . . .                     | 98  |
| 2.6.324 Falsifiability via Predictive Processing . . . . .                | 98  |
| 2.6.325 Falsifiability via Comparison . . . . .                           | 98  |
| 2.6.326 Falsifiability via Memory . . . . .                               | 98  |
| 2.6.327 Falsifiability via Cognitive Flows J_6J_6 . . . . .               | 99  |
| 2.6.328 Falsifiability via Structural Tension T_6T_6 . . . . .            | 99  |
| 2.6.329 Falsifiability of Transition K_5 K_6K_5 K_6 . . . . .             | 99  |
| 2.6.330 Falsifiability of K_7K_7 . . . . .                                | 99  |
| 2.6.331 Falsifiability via Communication Structure . . . . .              | 99  |
| 2.6.332 Falsifiability via Trust . . . . .                                | 100 |
| 2.6.333 Falsifiability via Roles . . . . .                                | 100 |
| 2.6.334 Falsifiability via Coordination Dynamics . . . . .                | 100 |
| 2.6.335 Falsifiability via Resource Flows J_7J_7 . . . . .                | 100 |
| 2.6.336 Falsifiability via Structural Tension T_7T_7 . . . . .            | 100 |
| 2.6.337 Falsifiability of Transition K_6 K_7K_6 K_7 . . . . .             | 100 |
| 2.6.338 Falsifiability of K_8K_8 . . . . .                                | 101 |
| 2.6.339 Falsifiability via Technology . . . . .                           | 101 |
| 2.6.340 Falsifiability via Cultural Dynamics . . . . .                    | 101 |
| 2.6.341 Falsifiability via Civilisational Flows J_8J_8 . . . . .          | 101 |
| 2.6.342 Falsifiability via Structural Tension T_8T_8 . . . . .            | 101 |
| 2.6.343 Falsifiability of Transition K_7 K_8K_7 K_8 . . . . .             | 102 |
| 2.6.344 Falsifiability of K_9K_9 . . . . .                                | 102 |
| 2.6.345 Falsifiability via Coherence . . . . .                            | 102 |
| 2.6.346 Falsifiability via Paradigm Dynamics . . . . .                    | 102 |
| 2.6.347 Falsifiability via Epistemic Potentials . . . . .                 | 102 |
| 2.6.348 Falsifiability via Methodological Structure . . . . .             | 102 |
| 2.6.349 Falsifiability via Scientific Flows J_9J_9 . . . . .              | 103 |
| 2.6.350 Falsifiability via Structural Tension T_9T_9 . . . . .            | 103 |
| 2.6.351 Falsifiability of the Transition K_8 K_9K_8 K_9 . . . . .         | 103 |
| 2.6.352 Falsifiability . . . . .                                          | 103 |
| 2.6.353 Structural Meaning of Falsifiability . . . . .                    | 103 |
| 2.6.354 Universal Falsifiability Criteria . . . . .                       | 104 |
| 2.6.355 Cross-Level Falsifiability Structure . . . . .                    | 104 |
| 2.6.356 Falsifiability via Embedding Spaces . . . . .                     | 104 |
| 2.6.357 Falsifiability in Practice: Multi-Domain Implementation . . . . . | 104 |
| 2.7 Source Trace . . . . .                                                | 104 |

# 1 OC Core Reviewer Attack and Response Map v1.3.3

## 1.1 Publication Identity

**Author.** Alexander Yashin, Independent Researcher, ORCID 0009-0008-6166-0914. **Research instrument.** Logion is the research-instrument and institute-automation system used to prepare, check, package, and audit the work; it is not an author. **Methodological framework.** ESTRa is the methodological framework used in the work; it is not an author or affiliation. **Version.** 1.3.3. **Release identity.** oc\_core\_1\_3\_3. **Concept DOI.** 10.5281/zenodo.17899134. **Artifact role.**

Adversarial objections, boundaries, and response map. **Intended audience.** hostile reviewers testing novelty, boundaries, evidence support, and reopening conditions.

## 1.2 Dedication

Dedicated to my dear wife Maria, without whom this work would have been impossible.

## 1.3 Acknowledgements

Substantive review and idea acknowledgements. They are acknowledged for review pressure, ideas, criticism, or external response that improved the work. Acknowledgement does not imply authorship, endorsement, publication approval, or agreement with the theory's release form.

- G. V. Apostolov
- Eduard Fadeev
- Gennady Alekseevich Nosov
- Sergey Shpadyrev
- Stanislav Tsukrov

## 1.4 Abstract

This map presents the release under hostile review. It groups objections by attacked claim, explains why each attack matters, names the response route, and records residual risk and reopening conditions.

## 1.5 Reader Contract

Read this document by choosing the claim under attack, then following objection, response, evidence, residual risk, and reopening condition. The release promotes evidence-bound model-core claims and excludes unsupported complete-science closure, unrestricted all-domain numerical completion, or unrestricted superiority-over-modern-science claims.

## 1.6 Table of Contents

The generated PDF contents provide page locations. The reader-facing route is:

1. Publication identity
2. Dedication
3. Acknowledgements
4. Abstract
5. Reader contract
6. Reading map
7. Block 2: Define Reader Contract
8. Block 2: Define Intended Audiences
9. Block 2: Define What OC Core Claims
10. Block 2: Define What OC Core Does Not Claim
11. Block 2: Define Release Scope vs Full Science Program
12. Block 2: Define Claim Promotion, Demotion,
13. Block 13: Define Falsifier Registry
14. Block 13: Define Counterexample Search

15. Block 13: Define Known Failure Modes
16. Block 13: Define Unsupported Strong Claims
17. Block 13: Define Demotion Rules
18. Block 13: Define Residual Scientific Risks
19. Block 13: Define Full-Science Background Obligations
20. Block 14: Define Novelty Criteria
21. Block 14: Define Non-Equivalence Criteria
22. Block 14: Define Overlap With Prior Art
23. Block 14: Define Residual-Delta Analysis
24. Block 14: Define Competitor-by-Competitor Comparison
25. Block 14: Define Response to “This Is Just Another Theory”
26. Block 14: Define Positioning of OC Core
27. Block 16: Define Review Protocol
28. Block 16: Define Cerberus Role Map
29. Block 16: Define Critical
30. Block 16: Define Closure Evidence
31. Block 16: Define Claim Corrections
32. Block 16: Define Hostile Reviewer Objections
33. Block 16: Define Response Matrix
34. Block 16: Define Remaining Non-Blocking Caveats
35. Block 18: Define Versioning
36. Block 18: Define Public Release Asset Set
37. Block 18: Define Journal Package Map
38. Block 18: Define Citation
39. Block 18: Define Publication Boundary
40. Block 18: Define Owner Approval Boundary
41. Additional sections continue in document order.

## 2 Body

### 2.1 Block 2

#### 2.1.1 Define Reader Contract

Define Reader Contract is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Reader Contract. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Reader Contract to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable

when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Reader Contract states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Reader Contract synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### **2.1.2 Define Intended Audiences**

Define Intended Audiences and Reading Paths is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Intended Audiences and Reading Paths. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Intended Audiences and Reading Paths to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Intended Audiences and Reading Paths states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Intended Audiences and Reading Paths synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### 2.1.3 Define What OC Core Claims

Define What OC Core Claims is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for What OC Core Claims. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind What OC Core Claims to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for What OC Core Claims states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize What OC Core Claims synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### 2.1.4 Define What OC Core Does Not Claim

Define What OC Core Does Not Claim is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for What OC Core Does Not Claim. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind What OC Core Does Not Claim to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited

proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for What OC Core Does Not Claim states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize What OC Core Does Not Claim synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### **2.1.5 Define Release Scope vs Full Science Program**

Define Release Scope vs Full Science Program is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition\_model route for Release Scope vs Full Science Program. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Release Scope vs Full Science Program to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Release Scope vs Full Science Program states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Release Scope vs Full Science Program synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next

obligation begins by defining the next object before asking the reader to accept claims about it.

### **2.1.6 Define Claim Promotion, Demotion,**

Define Claim Promotion, Demotion, and Background Obligations is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Claim Promotion, Demotion, and Background Obligations. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Claim Promotion, Demotion, and Background Obligations to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Claim Promotion, Demotion, and Background Obligations states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Claim Promotion, Demotion, and Background Obligations synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

## **2.2 Block 13**

### **2.2.1 Define Falsifier Registry**

Define Falsifier Registry is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Falsifier Registry. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.



Bind Falsifier Registry to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Falsifier Registry states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Falsifier Registry synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

## **2.2.2 Define Counterexample Search**

Define Counterexample Search is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Counterexample Search. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Counterexample Search to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Counterexample Search states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Counterexample Search synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### 2.2.3 Define Known Failure Modes

Define Known Failure Modes is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Known Failure Modes. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Known Failure Modes to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Known Failure Modes states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Known Failure Modes synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### 2.2.4 Define Unsupported Strong Claims

Define Unsupported Strong Claims is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Unsupported Strong Claims. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Unsupported Strong Claims to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Unsupported Strong Claims states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Unsupported Strong Claims synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### 2.2.5 Define Demotion Rules

Define Demotion Rules is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Demotion Rules. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Demotion Rules to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Demotion Rules states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Demotion Rules synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### **2.2.6 Define Residual Scientific Risks**

Define Residual Scientific Risks is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Residual Scientific Risks. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Residual Scientific Risks to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Residual Scientific Risks states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Residual Scientific Risks synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### **2.2.7 Define Full-Science Background Obligations**

Define Full-Science Background Obligations is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Full-Science Background Obligations. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Full-Science Background Obligations to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Full-Science Background Obligations states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Full-Science Background Obligations synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

## **2.3 Block 14**

### **2.3.1 Define Novelty Criteria**

Define Novelty Criteria is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Novelty Criteria. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Novelty Criteria to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Novelty Criteria states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background

research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Novelty Criteria synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### **2.3.2 Define Non-Equivalence Criteria**

Define Non-Equivalence Criteria is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Non-Equivalence Criteria. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Non-Equivalence Criteria to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Non-Equivalence Criteria states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Non-Equivalence Criteria synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### **2.3.3 Define Overlap With Prior Art**

Define Overlap With Prior Art is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Overlap With Prior Art. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary

is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Overlap With Prior Art to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Overlap With Prior Art states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Overlap With Prior Art synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### **2.3.4 Define Residual-Delta Analysis**

Define Residual-Delta Analysis is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Residual-Delta Analysis. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Residual-Delta Analysis to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Residual-Delta Analysis states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or

residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Residual-Delta Analysis synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### **2.3.5 Define Competitor-by-Competitor Comparison**

Define Competitor-by-Competitor Comparison is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition\_model route for Competitor-by-Competitor Comparison. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Competitor-by-Competitor Comparison to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Competitor-by-Competitor Comparison states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Competitor-by-Competitor Comparison synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### **2.3.6 Define Response to “This Is Just Another Theory”**

Define Response to “This Is Just Another Theory” is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the



definition\_model route for Response to “This Is Just Another Theory”. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Response to “This Is Just Another Theory” to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Response to “This Is Just Another Theory” states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Response to “This Is Just Another Theory” synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### **2.3.7 Define Positioning of OC Core**

Define Positioning of OC Core is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition\_model route for Positioning of OC Core. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Positioning of OC Core to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Positioning of OC Core states what would make the local claim

weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Positioning of OC Core synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

## **2.4 Block 16**

### **2.4.1 Define Review Protocol**

Define Review Protocol is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition model route for Review Protocol. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Review Protocol to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Review Protocol states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Review Protocol synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### 2.4.2 Define Cerberus Role Map

Define Cerberus Role Map is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Cerberus Role Map. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Cerberus Role Map to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Cerberus Role Map states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Cerberus Role Map synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### 2.4.3 Define Critical

Define Critical and High Findings is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Critical and High Findings. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Critical and High Findings to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited

proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Critical and High Findings states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Critical and High Findings synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

#### **2.4.4 Define Closure Evidence**

Define Closure Evidence is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Closure Evidence. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Closure Evidence to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Closure Evidence states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Closure Evidence synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by

defining the next object before asking the reader to accept claims about it.

### **2.4.5 Define Claim Corrections**

Define Claim Corrections is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Claim Corrections. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Claim Corrections to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Claim Corrections states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Claim Corrections synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### **2.4.6 Define Hostile Reviewer Objections**

Define Hostile Reviewer Objections is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Hostile Reviewer Objections. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Hostile Reviewer Objections to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact

paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Hostile Reviewer Objections states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Hostile Reviewer Objections synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

#### **2.4.7 Define Response Matrix**

Define Response Matrix is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Response Matrix. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Response Matrix to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Response Matrix states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Response Matrix synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new

scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### **2.4.8 Define Remaining Non-Blocking Caveats**

Define Remaining Non-Blocking Caveats is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Remaining Non-Blocking Caveats. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Remaining Non-Blocking Caveats to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Remaining Non-Blocking Caveats states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Remaining Non-Blocking Caveats synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

## **2.5 Block 18**

### **2.5.1 Define Versioning**

Define Versioning and Release Identity is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Versioning and Release Identity. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Versioning and Release Identity to proof, evidence, or replay carries the evidential burden for

the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Versioning and Release Identity states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Versioning and Release Identity synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### **2.5.2 Define Public Release Asset Set**

Define Public Release Asset Set is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Public Release Asset Set. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Public Release Asset Set to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Public Release Asset Set states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.



Synthesize Public Release Asset Set synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### 2.5.3 Define Journal Package Map

Define Journal Package Map is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Journal Package Map. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Journal Package Map to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Journal Package Map states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Journal Package Map synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### 2.5.4 Define Citation

Define Citation and Metadata Governance is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Citation and Metadata Governance. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Citation and Metadata Governance to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Citation and Metadata Governance states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Citation and Metadata Governance synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### **2.5.5 Define Publication Boundary**

Define Publication Boundary is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Publication Boundary. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Publication Boundary to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Publication Boundary states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Publication Boundary synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### **2.5.6 Define Owner Approval Boundary**

Define Owner Approval Boundary is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Owner Approval Boundary. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Owner Approval Boundary to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Owner Approval Boundary states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Owner Approval Boundary synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### **2.5.7 Define Incident**

Define Incident and Known-Error Lessons is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Incident and Known-Error Lessons. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Incident and Known-Error Lessons to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Incident and Known-Error Lessons states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Incident and Known-Error Lessons synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

### **2.5.8 Define External Use Guidance**

Define External Use Guidance is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for External Use Guidance. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind External Use Guidance to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for External Use Guidance states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize External Use Guidance synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

## **2.6 Block 999**

### **2.6.1 Adversarial Review Structure**

Adversarial Review Structure and Audience states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.2 Limits of recovery**

Limits of recovery states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.3 K-level criteria**

K-level criteria and falsifiers after external review states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.4 Falsifiability of K\_0K\_0**

Falsifiability of K\_0K\_0 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.5 Fundamental Falsifiability Principles for K\_0K\_0**

Fundamental Falsifiability Principles for K\_0K\_0 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk

carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.6 Structural Falsifiability Conditions**

Structural Falsifiability Conditions states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.7 Boundary**

Boundary and Region Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.8 Energy Functional**

Energy Functional and Threshold Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.9 Cycle**

Cycle and Evolution Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.10 Falsifiability via Modal Spaces**

Falsifiability via Modal Spaces and Modal Dimensions states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.11 Falsifiability via Meta-Cycles C\_10**

Falsifiability via Meta-Cycles C\_10 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.12 Falsifiability via Cycles C\_11**

Falsifiability via Cycles C\_11 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.13 Falsifiability via Universal Cycles C\_12**

Falsifiability via Universal Cycles C\_12 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.14 Percolation**

Percolation and Threshold Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.15 Boundary**

Boundary and Admissibility Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.16 Falsifiability via Chemical Potentials**

Falsifiability via Chemical Potentials and Thresholds states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk

carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.17 Falsifiability via Cycles C\_4C\_4**

Falsifiability via Cycles C\_4C\_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.18 Falsifiability via Membrane**

Falsifiability via Membrane and Electrical Boundary states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.19 Falsifiability via Neural Cycles C\_5C\_5**

Falsifiability via Neural Cycles C\_5C\_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.20 Falsifiability via Cognitive Cycles C\_6C\_6**

Falsifiability via Cognitive Cycles C\_6C\_6 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.21 Falsifiability via Social Cycles C\_7C\_7**

Falsifiability via Social Cycles C\_7C\_7 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.



### **2.6.22 Falsifiability via Civilisational Cycles C\_8C\_8**

Falsifiability via Civilisational Cycles C\_8C\_8 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.23 Falsifiability via Scientific Cycles C\_9C\_9**

Falsifiability via Scientific Cycles C\_9C\_9 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.24 Falsifiability via Dynamics**

Falsifiability via Dynamics and Collapse states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.25 Falsifier**

Falsifier and collapse boundary states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.26 Theorem fate correction after external review**

Theorem fate correction after external review carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.27 Worked Examples, Counterexamples,**

Worked Examples, Counterexamples, and Reader-Oriented Interpretive Checks carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose

to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.28 What would count as a real counterexample**

What would count as a real counterexample carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.29 Mathematics: Strict closure**

Mathematics: Strict closure and counterexample playbook carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.30 Simulation**

Simulation and Counterexample Search carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.31 Reviewer challenge 3: Minimality Component Witness**

Reviewer challenge 3: Minimality Component Witness carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.32 Reviewer challenge 4: Klevel Transition Witness**

Reviewer challenge 4: Klevel Transition Witness carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.33 Falsifiability via Symbolic Systems**

Falsifiability via Symbolic Systems and Writing carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.34 Falsifiability via Knowledge Systems**

Falsifiability via Knowledge Systems and Epistemic Structure carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.35 (P10.11) Completeness Is Structurally Limited**

(P10.11) Completeness Is Structurally Limited carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.36 (i) Structural falsifiability**

- (i) Structural falsifiability carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.37 Replay Matrix for Independent Review**

Replay Matrix for Independent Review carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.38 Reviewer challenge 5: Numeric Replay Quarantine**

Reviewer challenge 5: Numeric Replay Quarantine carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.39 Attack Family 3: The Empirical Rows Are Too Weak**

Attack Family 3: The Empirical Rows Are Too Weak carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.40 Formal-Mathematics Reviewer**

Formal-Mathematics Reviewer carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.41 Empirical-Statistics Reviewer**

Empirical-Statistics Reviewer carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.42 Publication-Metadata Reviewer**

Publication-Metadata Reviewer carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.43 Worked Attack Transcript B: Replay Surface**

Worked Attack Transcript B: Replay Surface carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.44 Reviewer challenge 7: Prior Art Positioning**

Reviewer challenge 7: Prior Art Positioning states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.45 Worked Attack Transcript C: Prior-Art Surface**

Worked Attack Transcript C: Prior-Art Surface states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.46 Phenomenon Coverage**

Phenomenon Coverage and Limits states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.47 Reviewer challenge 6: Phenomenon Coverage Model Card**

Reviewer challenge 6: Phenomenon Coverage Model Card states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.48 Reviewer challenge 8: Attack Matrix Distinct Surface Coverage**

Reviewer challenge 8: Attack Matrix Distinct Surface Coverage states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.49 Attack Family 4: Phenomenon Coverage Is Inflated**

Attack Family 4: Phenomenon Coverage Is Inflated states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.50 Visual Route - Reviewer Route Figures**

Visual Route - Reviewer Route Figures states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.51 Reviewer Navigation Matrix**

Reviewer Navigation Matrix and Audit Checklist states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.52 Review questions, compact route,**

Review questions, compact route, and decisive checks states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.53 Audit, provenance,**

Audit, provenance, and review posture states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.54 Compact reading order by reviewer role**

Compact reading order by reviewer role states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.55 Rapid critical reviewer**

Rapid critical reviewer states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.56 Formal mathematical reviewer**

Formal mathematical reviewer states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.57 Generalist scientific reviewer**

Generalist scientific reviewer states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It

must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.58 Reviewer note**

Reviewer note states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.59 Tuple**

Tuple and hierarchy limits states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.60 Reviewer acknowledgement**

Reviewer acknowledgement states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.61 Hostile-review backlog**

Hostile-review backlog states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.62 Shortest hostile-review checklist**

Shortest hostile-review checklist states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.



### **2.6.63 Reviewer Entry Points**

Reviewer Entry Points states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.64 Does OC actually explain minimality versus relabeling attack?**

Does OC actually explain minimality versus relabeling attack? states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.65 Reviewer Checklist**

Reviewer Checklist states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.66 Reviewer Burden**

Reviewer Burden and Author Burden states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.67 Reviewer Response Method**

Reviewer Response Method states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.68 Claim Boundary**

Claim Boundary and Research Limits states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate

scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.69 Reviewer challenge 1: Theorem Proof Binding**

Reviewer challenge 1: Theorem Proof Binding carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.70 Reviewer challenge 2: Theorem Public Surface Binding**

Reviewer challenge 2: Theorem Public Surface Binding carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

#### **2.6.71 Adversarial Review Method**

Adversarial Review Method states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.72 Attack Family 1: The Theory Is Only a Reframing**

Attack Family 1: The Theory Is Only a Reframing states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.73 Attack Family 2: The Formal Claims Are Theatre**

Attack Family 2: The Formal Claims Are Theatre states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure

and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.74 Attack Family 5: Public Wording Outruns Evidence**

Attack Family 5: Public Wording Outruns Evidence carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

#### **2.6.75 Attack Family 6: The Package Is Not a Scientific Text**

Attack Family 6: The Package Is Not a Scientific Text states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.76 Reviewer Role Playbook**

Reviewer Role Playbook states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.77 Computational-Semantics Reviewer**

Computational-Semantics Reviewer states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.78 Release-Engineering Reviewer**

Release-Engineering Reviewer states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.79 Synthesis Reviewer**

Synthesis Reviewer states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.80 Minimum Hostile Review Walkthrough**

Minimum Hostile Review Walkthrough states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.81 End-to-End Attack Drill**

End-to-End Attack Drill states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.82 Cheap Review Cascade**

Cheap Review Cascade states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.83 Worked Attack Transcript A: Theorem Surface**

Worked Attack Transcript A: Theorem Surface carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

### **2.6.84 Worked Attack Transcript D: Public Surface**

Worked Attack Transcript D: Public Surface states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to

overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.85 Worked Attack Transcript E: Synthesis Surface**

Worked Attack Transcript E: Synthesis Surface states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.86 Falsifiability via Institutions**

Falsifiability via Institutions and Governance states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.87 Limitations of the current formalism**

Limitations of the current formalism states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.88 Limits of recovery**

Limits of recovery states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.89 Extended Falsifiability**

Extended Falsifiability and Experimental Programme states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.90 Meaning of Falsifiability for Structural Theories**

Meaning of Falsifiability for Structural Theories states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.91 Structural vs phenomenological falsifiability**

Structural vs phenomenological falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.92 Structural falsification channels**

Structural falsification channels states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.93 Cross-domain falsification logic**

Cross-domain falsification logic states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.94 Falsification criteria**

Falsification criteria and tests states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.95 Source-tradition remediation after external review**

Source-tradition remediation after external review states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk

carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.96 K-level criteria**

K-level criteria and falsifiers after external review states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.97 Theorem fate correction after external review**

Theorem fate correction after external review carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.98 Meta-limit extension**

Meta-limit extension states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.99 Type~VII Experiments: Detectability**

Type~VII Experiments: Detectability and Falsification states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.100 Relation to Falsifiability**

Relation to Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.101 Falsifiability of K\_0K\_0**

Falsifiability of K\_0K\_0 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.102 Fundamental Falsifiability Principles for K\_0K\_0**

Fundamental Falsifiability Principles for K\_0K\_0 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.103 Structural Falsifiability Conditions**

Structural Falsifiability Conditions states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.104 Boundary**

Boundary and Region Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.105 Falsifiability of K\_1K\_1**

Falsifiability of K\_1K\_1 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.106 Structural**

Structural and Topological Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk



carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.107 Energy Functional**

Energy Functional and Threshold Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.108 Cycle**

Cycle and Evolution Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.109 Embedding-Space**

Embedding-Space and Transition Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.110 Falsifiability of K<sub>10</sub>**

Falsifiability of K<sub>10</sub> states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.111 Falsifiability via Reflexive**

Falsifiability via Reflexive and Meta-Theoretical Coherence states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.112 Falsifiability via Category-Theoretical**

Falsifiability via Category-Theoretical and Functorial Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.113 Falsifiability via Modal Spaces**

Falsifiability via Modal Spaces and Modal Dimensions states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.114 Falsifiability via Meta-Level Potentials P<sub>10</sub>**

Falsifiability via Meta-Level Potentials P<sub>10</sub> states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.115 Falsifiability via Meta-Flows J<sup>^(10)</sup>**

Falsifiability via Meta-Flows J<sup>^(10)</sup> states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.116 Falsifiability via Meta-Cycles C<sub>10</sub>**

Falsifiability via Meta-Cycles C<sub>10</sub> states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.117 Falsifiability via Structural Tension T\_10**

Falsifiability via Structural Tension T\_10 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.118 Falsifiability of the Transition K\_9 K\_10**

Falsifiability of the Transition K\_9 K\_10 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.119 Falsifiability of K\_11**

Falsifiability of K\_11 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.120 Falsifiability via Meta-Meta Coherence**

Falsifiability via Meta-Meta Coherence states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.121 Falsifiability via Higher-Order Categorical Structures**

Falsifiability via Higher-Order Categorical Structures states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.122 Falsifiability via Higher-Order Modal Spaces**

Falsifiability via Higher-Order Modal Spaces states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to

overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.123 Falsifiability via Potentials P\_11**

Falsifiability via Potentials P\_11 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.124 Falsifiability via Flows $\hat{J}(11)$**

Falsifiability via Flows  $\hat{J}(11)$  states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.125 Falsifiability via Cycles C\_11**

Falsifiability via Cycles C\_11 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.126 Falsifiability via Structural Tension T\_11**

Falsifiability via Structural Tension T\_11 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.127 Falsifiability of the Transition K\_10 K\_11**

Falsifiability of the Transition K\_10 K\_11 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.128 Falsifiability of K\_12**

Falsifiability of K\_12 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.129 Falsifiability via Universal Structural Consistency**

Falsifiability via Universal Structural Consistency states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.130 Falsifiability via Universal Modal**

Falsifiability via Universal Modal and Categorical Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.131 Falsifiability via Universal Potentials P\_12**

Falsifiability via Universal Potentials P\_12 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.132 Falsifiability via Universal Flows $J^{(12)}$**

Falsifiability via Universal Flows  $J^{(12)}$  states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.133 Falsifiability via Universal Cycles C\_12**

Falsifiability via Universal Cycles C\_12 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to

overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.134 Falsifiability via Universal Boundaries (K\_12)**

Falsifiability via Universal Boundaries (K\_12) states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.135 Falsifiability of K\_11 K\_12 Transition**

Falsifiability of K\_11 K\_12 Transition states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.136 Falsifiability of K\_2K\_2**

Falsifiability of K\_2K\_2 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.137 Topological**

Topological and Connectivity Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.138 Percolation**

Percolation and Threshold Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.139 Boundary**

Boundary and Admissibility Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.140 Potentials, Flows**

Potentials, Flows and Field Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.141 Dynamic**

Dynamic and Evolutionary Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.142 Transition**

Transition and Embedding Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.143 Falsifiability of K\_3K\_3**

Falsifiability of K\_3K\_3 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.144 Falsifiability via Chemical Configuration**

Falsifiability via Chemical Configuration states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk

carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.145 Falsifiability via Reaction Networks**

Falsifiability via Reaction Networks states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.146 Falsifiability via RAF Networks**

Falsifiability via RAF Networks and Closure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.147 Falsifiability via Chemical Potentials**

Falsifiability via Chemical Potentials and Thresholds states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.148 Falsifiability via Boundaries**

Falsifiability via Boundaries and Admissibility states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.149 Falsifiability via Dynamics**

Falsifiability via Dynamics and Evolution states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.



#### **2.6.150 Falsifiability via Transition K\_2 K\_3K\_2 K\_3**

Falsifiability via Transition K\_2 K\_3K\_2 K\_3 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.151 Falsifiability of K\_4K\_4**

Falsifiability of K\_4K\_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.152 Falsifiability via Compartment Formation**

Falsifiability via Compartment Formation states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.153 Falsifiability via Biological Potentials P\_4P\_4**

Falsifiability via Biological Potentials P\_4P\_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.154 Falsifiability via Flows J\_4J\_4**

Falsifiability via Flows J\_4J\_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.155 Falsifiability via Cycles C\_4C\_4**

Falsifiability via Cycles C\_4C\_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical

toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.156 Falsifiability via Membrane-Bound Information**

Falsifiability via Membrane-Bound Information and Regulation states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.157 Falsifiability via Structural Tension T\_4T\_4**

Falsifiability via Structural Tension T\_4T\_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.158 Falsifiability via Transition K\_3 K\_4K\_3 K\_4**

Falsifiability via Transition K\_3 K\_4K\_3 K\_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.159 Falsifiability of K\_5K\_5**

Falsifiability of K\_5K\_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.160 Falsifiability via Membrane**

Falsifiability via Membrane and Electrical Boundary states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.161 Falsifiability via Electrical Potentials P\_5P\_5**

Falsifiability via Electrical Potentials P\_5P\_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.162 Falsifiability via Ion Flows J\_5J\_5**

Falsifiability via Ion Flows J\_5J\_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.163 Falsifiability via Neural Cycles C\_5C\_5**

Falsifiability via Neural Cycles C\_5C\_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.164 Falsifiability via Stochastic Logic**

Falsifiability via Stochastic Logic and Information Flow states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.165 Falsifiability via Structural Tension T\_5T\_5**

Falsifiability via Structural Tension T\_5T\_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.166 Falsifiability of Transition K\_4 K\_5K\_4 K\_5**

Falsifiability of Transition K\_4 K\_5K\_4 K\_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission

to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.167 Falsifiability of K\_6K\_6**

Falsifiability of K\_6K\_6 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.168 Falsifiability via Representational Structure**

Falsifiability via Representational Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.169 Falsifiability via Binding Dynamics**

Falsifiability via Binding Dynamics states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.170 Falsifiability via Predictive Processing**

Falsifiability via Predictive Processing states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.171 Falsifiability via Comparison**

Falsifiability via Comparison and Selection states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.172 Falsifiability via Memory**

Falsifiability via Memory and Stability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.173 Falsifiability via Cognitive Flows J\_6J\_6**

Falsifiability via Cognitive Flows J\_6J\_6 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.174 Falsifiability via Cognitive Cycles C\_6C\_6**

Falsifiability via Cognitive Cycles C\_6C\_6 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.175 Falsifiability via Structural Tension T\_6T\_6**

Falsifiability via Structural Tension T\_6T\_6 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.176 Falsifiability of Transition K\_5 K\_6K\_5 K\_6**

Falsifiability of Transition K\_5 K\_6K\_5 K\_6 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.177 Falsifiability of K\_7K\_7**

Falsifiability of K\_7K\_7 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical

toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.178 Falsifiability via Communication Structure**

Falsifiability via Communication Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.179 Falsifiability via Trust**

Falsifiability via Trust and Cohesion states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.180 Falsifiability via Roles**

Falsifiability via Roles and Normative Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.181 Falsifiability via Coordination Dynamics**

Falsifiability via Coordination Dynamics states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.182 Falsifiability via Resource Flows J\_7J\_7**

Falsifiability via Resource Flows J\_7J\_7 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.183 Falsifiability via Social Cycles C\_7C\_7**

Falsifiability via Social Cycles C\_7C\_7 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.184 Falsifiability via Structural Tension T\_7T\_7**

Falsifiability via Structural Tension T\_7T\_7 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.185 Falsifiability of Transition K\_6 K\_7K\_6 K\_7**

Falsifiability of Transition K\_6 K\_7K\_6 K\_7 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.186 Falsifiability of K\_8K\_8**

Falsifiability of K\_8K\_8 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.187 Falsifiability via Technology**

Falsifiability via Technology and Infrastructure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.188 Falsifiability via Cultural Dynamics**

Falsifiability via Cultural Dynamics states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried

by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.189 Falsifiability via Civilisational Flows J\_8J\_8**

Falsifiability via Civilisational Flows J\_8J\_8 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.190 Falsifiability via Civilisational Cycles C\_8C\_8**

Falsifiability via Civilisational Cycles C\_8C\_8 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.191 Falsifiability via Structural Tension T\_8T\_8**

Falsifiability via Structural Tension T\_8T\_8 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.192 Falsifiability of Transition K\_7 K\_8K\_7 K\_8**

Falsifiability of Transition K\_7 K\_8K\_7 K\_8 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.193 Falsifiability of K\_9K\_9**

Falsifiability of K\_9K\_9 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.



#### **2.6.194 Falsifiability via Coherence**

Falsifiability via Coherence and Logical Consistency states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.195 Falsifiability via Paradigm Dynamics**

Falsifiability via Paradigm Dynamics states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.196 Falsifiability via Epistemic Potentials**

Falsifiability via Epistemic Potentials and Explanatory Power states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.197 Falsifiability via Methodological Structure**

Falsifiability via Methodological Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.198 Falsifiability via Scientific Flows J\_9J\_9**

Falsifiability via Scientific Flows J\_9J\_9 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.199 Falsifiability via Scientific Cycles C\_9C\_9**

Falsifiability via Scientific Cycles C\_9C\_9 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to

overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.200 Falsifiability via Structural Tension T\_9T\_9**

Falsifiability via Structural Tension T\_9T\_9 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.201 Falsifiability of the Transition K\_8 K\_9K\_8 K\_9**

Falsifiability of the Transition K\_8 K\_9K\_8 K\_9 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.202 Falsifiability**

Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.203 Structural Meaning of Falsifiability**

Structural Meaning of Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.204 Universal Falsifiability Criteria**

Universal Falsifiability Criteria states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.205 Cross-Level Falsifiability Structure**

Cross-Level Falsifiability Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.206 Falsifiability via Dynamics**

Falsifiability via Dynamics and Collapse states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.207 Falsifiability via Embedding Spaces**

Falsifiability via Embedding Spaces states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.208 Falsifiability in Practice: Multi-Domain Implementation**

Falsifiability in Practice: Multi-Domain Implementation states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.209 Falsifier**

Falsifier and collapse boundary states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

### **2.6.210 Claim Boundary**

Claim Boundary and Research Limits states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate

scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.211 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-FALSE**

FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.212 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-TRUE**

FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.213 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-FALSE**

FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.214 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-TRUE**

FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.215 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-FALSE**

FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.216 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-TRUE**

FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.217 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-FALSE**

FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.218 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-TRUE**

FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.219 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-FALSE**

FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use

the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.220 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-TRUE**

FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.221 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-FALSE**

FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.222 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-TRUE**

FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.223 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-FALSE**

FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.224 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-TRUE**

FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.225 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-FALSE**

FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.226 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-FALSE**

FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.227 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-TRUE**

FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.228 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-FALSE**

FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control,

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**2.6.229 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-TRUE**

FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

**2.6.230 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-FALSE**

FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

**2.6.231 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-TRUE**

FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

**2.6.232 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-FALSE**

FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.



#### **2.6.233 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-TRUE**

FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.234 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-FALSE**

FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.235 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-TRUE**

FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.236 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-FALSE**

FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.237 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-TRUE**

FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control,

reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.238 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-FALSE**

FM-T133-ID-TT-RESIDUE-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.239 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-TRUE**

FM-T133-ID-TT-RESIDUE-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.240 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-FALSE**

FM-T133-ID-TT-RESIDUE-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.241 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-FALSE**

FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.242 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-TRUE**

FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.243 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-FALSE**

FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.244 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-TRUE**

FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.245 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-FALSE**

FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.246 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-TRUE**

FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control,

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**2.6.247 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-FALSE**

FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

**2.6.248 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-TRUE**

FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

**2.6.249 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-FALSE**

FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

**2.6.250 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-TRUE**

FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

**2.6.251 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-FALSE**

FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

**2.6.252 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-TRUE**

FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

**2.6.253 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-FALSE**

FM-T133-ID-TT-REBIRTH-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

**2.6.254 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-TRUE**

FM-T133-ID-TT-REBIRTH-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

**2.6.255 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-FALSE**

FM-T133-ID-TT-REBIRTH-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control,

reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.256 PHYSICAL\_REVIEW\_RESEARCH**

PHYSICAL\_REVIEW\_RESEARCH states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.257 Minimum Independent Scientific Review Packet**

Minimum Independent Scientific Review Packet states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.258 Limitations**

Limitations states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

#### **2.6.259 Journal Owner-Review Packages**

Journal Owner-Review Packages states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.260 Limitations of the current formalism**

Limitations of the current formalism states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.261 Extended Falsifiability**

Extended Falsifiability and Experimental Programme states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.262 Meaning of Falsifiability for Structural Theories**

Meaning of Falsifiability for Structural Theories states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.263 Structural vs phenomenological falsifiability**

Structural vs phenomenological falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.264 Structural falsification channels**

Structural falsification channels states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.265 Cross-domain falsification logic**

Cross-domain falsification logic states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.266 Falsification criteria**

Falsification criteria and tests states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It

must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.267 Source-tradition remediation after external review**

Source-tradition remediation after external review states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.268 Meta-limit extension**

Meta-limit extension states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.269 Type~VII Experiments: Detectability**

Type~VII Experiments: Detectability and Falsification states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.270 Relation to Falsifiability**

Relation to Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.271 Falsifiability of K\_1K\_1**

Falsifiability of K\_1K\_1 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.



### **2.6.272 Structural**

Structural and Topological Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.273 Embedding-Space**

Embedding-Space and Transition Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.274 Falsifiability of K\_10**

Falsifiability of K\_10 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.275 Falsifiability via Reflexive**

Falsifiability via Reflexive and Meta-Theoretical Coherence states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.276 Falsifiability via Category-Theoretical**

Falsifiability via Category-Theoretical and Functorial Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.277 Falsifiability via Meta-Level Potentials P\_10**

Falsifiability via Meta-Level Potentials P\_10 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.278 Falsifiability via Meta-Flows J^(10)**

Falsifiability via Meta-Flows J^(10) states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.279 Falsifiability via Structural Tension T\_10**

Falsifiability via Structural Tension T\_10 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.280 Falsifiability of the Transition K\_9 K\_10**

Falsifiability of the Transition K\_9 K\_10 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.281 Falsifiability of K\_11**

Falsifiability of K\_11 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.282 Falsifiability via Meta-Meta Coherence**

Falsifiability via Meta-Meta Coherence states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried

by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.283 Falsifiability via Higher-Order Categorical Structures**

Falsifiability via Higher-Order Categorical Structures states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.284 Falsifiability via Higher-Order Modal Spaces**

Falsifiability via Higher-Order Modal Spaces states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.285 Falsifiability via Potentials P<sub>11</sub>**

Falsifiability via Potentials P<sub>11</sub> states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.286 Falsifiability via Flows $J^{\wedge}(11)$**

Falsifiability via Flows  $J^{\wedge}(11)$  states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.287 Falsifiability via Structural Tension T<sub>11</sub>**

Falsifiability via Structural Tension T<sub>11</sub> states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.288 Falsifiability of the Transition K\_10 K\_11**

Falsifiability of the Transition K\_10 K\_11 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.289 Falsifiability of K\_12**

Falsifiability of K\_12 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.290 Falsifiability via Universal Structural Consistency**

Falsifiability via Universal Structural Consistency states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.291 Falsifiability via Universal Modal**

Falsifiability via Universal Modal and Categorical Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.292 Falsifiability via Universal Potentials P\_12**

Falsifiability via Universal Potentials P\_12 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.293 Falsifiability via Universal Flows $J^{\wedge}(12)$**

Falsifiability via Universal Flows  $J^{\wedge}(12)$  states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to

overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.294 Falsifiability via Universal Boundaries (K\_12)**

Falsifiability via Universal Boundaries (K\_12) states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.295 Falsifiability of K\_11 K\_12 Transition**

Falsifiability of K\_11 K\_12 Transition states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.296 Falsifiability of K\_2K\_2**

Falsifiability of K\_2K\_2 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.297 Topological**

Topological and Connectivity Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.298 Potentials, Flows**

Potentials, Flows and Field Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.299 Dynamic**

Dynamic and Evolutionary Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.300 Transition**

Transition and Embedding Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.301 Falsifiability of K\_3K\_3**

Falsifiability of K\_3K\_3 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.302 Falsifiability via Chemical Configuration**

Falsifiability via Chemical Configuration states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.303 Falsifiability via Reaction Networks**

Falsifiability via Reaction Networks states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.304 Falsifiability via RAF Networks**

Falsifiability via RAF Networks and Closure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk

carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.305 Falsifiability via Boundaries**

Falsifiability via Boundaries and Admissibility states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.306 Falsifiability via Dynamics**

Falsifiability via Dynamics and Evolution states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.307 Falsifiability via Transition K<sub>2</sub> K<sub>3</sub> K<sub>2</sub> K<sub>3</sub>**

Falsifiability via Transition K<sub>2</sub> K<sub>3</sub> K<sub>2</sub> K<sub>3</sub> states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.308 Falsifiability of K<sub>4</sub> K<sub>4</sub>**

Falsifiability of K<sub>4</sub> K<sub>4</sub> states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.309 Falsifiability via Compartment Formation**

Falsifiability via Compartment Formation states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.310 Falsifiability via Biological Potentials P\_4P\_4**

Falsifiability via Biological Potentials P\_4P\_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.311 Falsifiability via Flows J\_4J\_4**

Falsifiability via Flows J\_4J\_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.312 Falsifiability via Membrane-Bound Information**

Falsifiability via Membrane-Bound Information and Regulation states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.313 Falsifiability via Structural Tension T\_4T\_4**

Falsifiability via Structural Tension T\_4T\_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.314 Falsifiability via Transition K\_3 K\_4K\_3 K\_4**

Falsifiability via Transition K\_3 K\_4K\_3 K\_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.315 Falsifiability of K\_5K\_5**

Falsifiability of K\_5K\_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It



must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.316 Falsifiability via Electrical Potentials P\_5P\_5**

Falsifiability via Electrical Potentials P\_5P\_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.317 Falsifiability via Ion Flows J\_5J\_5**

Falsifiability via Ion Flows J\_5J\_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.318 Falsifiability via Stochastic Logic**

Falsifiability via Stochastic Logic and Information Flow states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.319 Falsifiability via Structural Tension T\_5T\_5**

Falsifiability via Structural Tension T\_5T\_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.320 Falsifiability of Transition K\_4 K\_5K\_4 K\_5**

Falsifiability of Transition K\_4 K\_5K\_4 K\_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.321 Falsifiability of K\_6K\_6**

Falsifiability of K\_6K\_6 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.322 Falsifiability via Representational Structure**

Falsifiability via Representational Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.323 Falsifiability via Binding Dynamics**

Falsifiability via Binding Dynamics states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.324 Falsifiability via Predictive Processing**

Falsifiability via Predictive Processing states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.325 Falsifiability via Comparison**

Falsifiability via Comparison and Selection states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.326 Falsifiability via Memory**

Falsifiability via Memory and Stability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried

by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.327 Falsifiability via Cognitive Flows J\_6J\_6**

Falsifiability via Cognitive Flows J\_6J\_6 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.328 Falsifiability via Structural Tension T\_6T\_6**

Falsifiability via Structural Tension T\_6T\_6 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.329 Falsifiability of Transition K\_5 K\_6K\_5 K\_6**

Falsifiability of Transition K\_5 K\_6K\_5 K\_6 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.330 Falsifiability of K\_7K\_7**

Falsifiability of K\_7K\_7 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.331 Falsifiability via Communication Structure**

Falsifiability via Communication Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.332 Falsifiability via Trust**

Falsifiability via Trust and Cohesion states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.333 Falsifiability via Roles**

Falsifiability via Roles and Normative Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.334 Falsifiability via Coordination Dynamics**

Falsifiability via Coordination Dynamics states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.335 Falsifiability via Resource Flows J\_7J\_7**

Falsifiability via Resource Flows J\_7J\_7 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.336 Falsifiability via Structural Tension T\_7T\_7**

Falsifiability via Structural Tension T\_7T\_7 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.337 Falsifiability of Transition K\_6 K\_7K\_6 K\_7**

Falsifiability of Transition K\_6 K\_7K\_6 K\_7 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk

carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.338 Falsifiability of K\_8K\_8**

Falsifiability of K\_8K\_8 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.339 Falsifiability via Technology**

Falsifiability via Technology and Infrastructure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.340 Falsifiability via Cultural Dynamics**

Falsifiability via Cultural Dynamics states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.341 Falsifiability via Civilisational Flows J\_8J\_8**

Falsifiability via Civilisational Flows J\_8J\_8 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.342 Falsifiability via Structural Tension T\_8T\_8**

Falsifiability via Structural Tension T\_8T\_8 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.343 Falsifiability of Transition K\_7 K\_8K\_7 K\_8**

Falsifiability of Transition K\_7 K\_8K\_7 K\_8 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.344 Falsifiability of K\_9K\_9**

Falsifiability of K\_9K\_9 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.345 Falsifiability via Coherence**

Falsifiability via Coherence and Logical Consistency states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.346 Falsifiability via Paradigm Dynamics**

Falsifiability via Paradigm Dynamics states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.347 Falsifiability via Epistemic Potentials**

Falsifiability via Epistemic Potentials and Explanatory Power states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.348 Falsifiability via Methodological Structure**

Falsifiability via Methodological Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to

overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.349 Falsifiability via Scientific Flows J\_9J\_9**

Falsifiability via Scientific Flows J\_9J\_9 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.350 Falsifiability via Structural Tension T\_9T\_9**

Falsifiability via Structural Tension T\_9T\_9 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.351 Falsifiability of the Transition K\_8 K\_9K\_8 K\_9**

Falsifiability of the Transition K\_8 K\_9K\_8 K\_9 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.352 Falsifiability**

Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

#### **2.6.353 Structural Meaning of Falsifiability**

Structural Meaning of Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.354 Universal Falsifiability Criteria**

Universal Falsifiability Criteria states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.355 Cross-Level Falsifiability Structure**

Cross-Level Falsifiability Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.356 Falsifiability via Embedding Spaces**

Falsifiability via Embedding Spaces states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

### **2.6.357 Falsifiability in Practice: Multi-Domain Implementation**

Falsifiability in Practice: Multi-Domain Implementation states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

## **2.7 Source Trace**

Exact source bindings, path hashes, quality scorer hooks, and transition records are recorded in the review package manifest. They are kept out of the main prose to avoid turning the document into a ledger dump.